

1. Explain Cloud Threat Detection

Cloud threat detection is the process of monitoring cloud environments to identify suspicious activities, security threats, and unauthorized access. As more organizations move their applications and data to cloud platforms such as **Amazon Web Services (AWS)**, **Microsoft Azure**, and **Google Cloud Platform (GCP)**, protecting these environments has become increasingly important.

Cloud threat detection uses security tools that continuously monitor user activity, network traffic, API calls, login attempts, and cloud resources. These tools help security teams quickly identify potential attacks and respond before they cause significant damage.

Common Cloud Threats

- Unauthorized account access.
- Stolen or compromised credentials.
- Misconfigured cloud storage.
- Malware or ransomware attacks.
- Insider threats.
- Data breaches.
- Privilege escalation.

Benefits of Cloud Threat Detection

- Early detection of security threats.
- Continuous monitoring of cloud resources.
- Faster incident response.
- Protection of sensitive data.
- Improved compliance with security standards.

2. Research IAM Abuse

Identity and Access Management (IAM) controls who can access cloud resources and what actions they are allowed to perform. IAM abuse occurs when attackers misuse user accounts, roles, or permissions to gain unauthorized access to cloud services.

Attackers often target IAM because compromising a single privileged account can provide access to multiple cloud resources.

Common IAM Abuse Techniques

1. Stolen Credentials

Attackers use phishing or malware to steal usernames and passwords and access cloud accounts.

2. Excessive Permissions

Users with unnecessary administrative privileges can accidentally or intentionally misuse their access.

3. Privilege Escalation

Attackers exploit misconfigured IAM policies to gain higher privileges.

4. Compromised API Keys

Exposed API keys stored in source code or public repositories can allow attackers to access cloud services.

5. Role Misconfiguration

Improperly configured IAM roles may allow unauthorized users to assume privileged roles.

Prevention

- Enable Multi-Factor Authentication (MFA).
- Apply the Principle of Least Privilege.
- Rotate passwords and API keys regularly.
- Monitor IAM activities continuously.
- Review IAM policies and permissions frequently.
- Disable unused accounts and credentials.

3. Analyze a Suspicious API Calls Scenario

Scenario

A company hosts its application on AWS. During routine monitoring, the SOC team notices that a user account is making an unusually high number of API calls late at night. The API requests include creating new IAM users, modifying security groups, and accessing sensitive storage buckets.

The user normally works during business hours and has never performed these actions before. The activity originates from an unfamiliar IP address in another country.

The cloud monitoring solution generates multiple alerts, and the SOC analyst begins an investigation.

Analysis

The unusual API activity suggests that the user's cloud credentials may have been compromised. The attacker appears to be attempting privilege escalation by creating new IAM users and changing security settings.

The analyst reviews the cloud audit logs and confirms that the API calls were made using the employee's access keys. The compromised credentials are immediately disabled, the user's password is reset, and Multi-Factor Authentication (MFA) is enforced. The SOC team also reviews all recent cloud activities to identify any additional unauthorized changes.

This scenario highlights the importance of monitoring API activity, as attackers often use cloud APIs to perform malicious actions without directly accessing the cloud management console.

Indicators of Compromise (IOCs)

- API calls from unfamiliar IP addresses.
- Login attempts from unusual geographic locations.
- Creation of unexpected IAM users or roles.
- Changes to IAM policies or permissions.
- Excessive API requests within a short period.
- Access to sensitive storage buckets outside normal working hours.
- Disabled security logging or monitoring services.
- Unusual privilege changes.

Response Steps

1. Investigate cloud audit logs (such as AWS CloudTrail or Azure Activity Logs).
2. Disable the compromised access keys or user account.
3. Reset passwords and enable Multi-Factor Authentication (MFA).
4. Review all recent API activities and configuration changes.
5. Remove unauthorized IAM users, roles, or policies.
6. Restore any modified cloud resources if necessary.
7. Monitor the environment for additional suspicious activity.
8. Document the incident and update security monitoring rules.